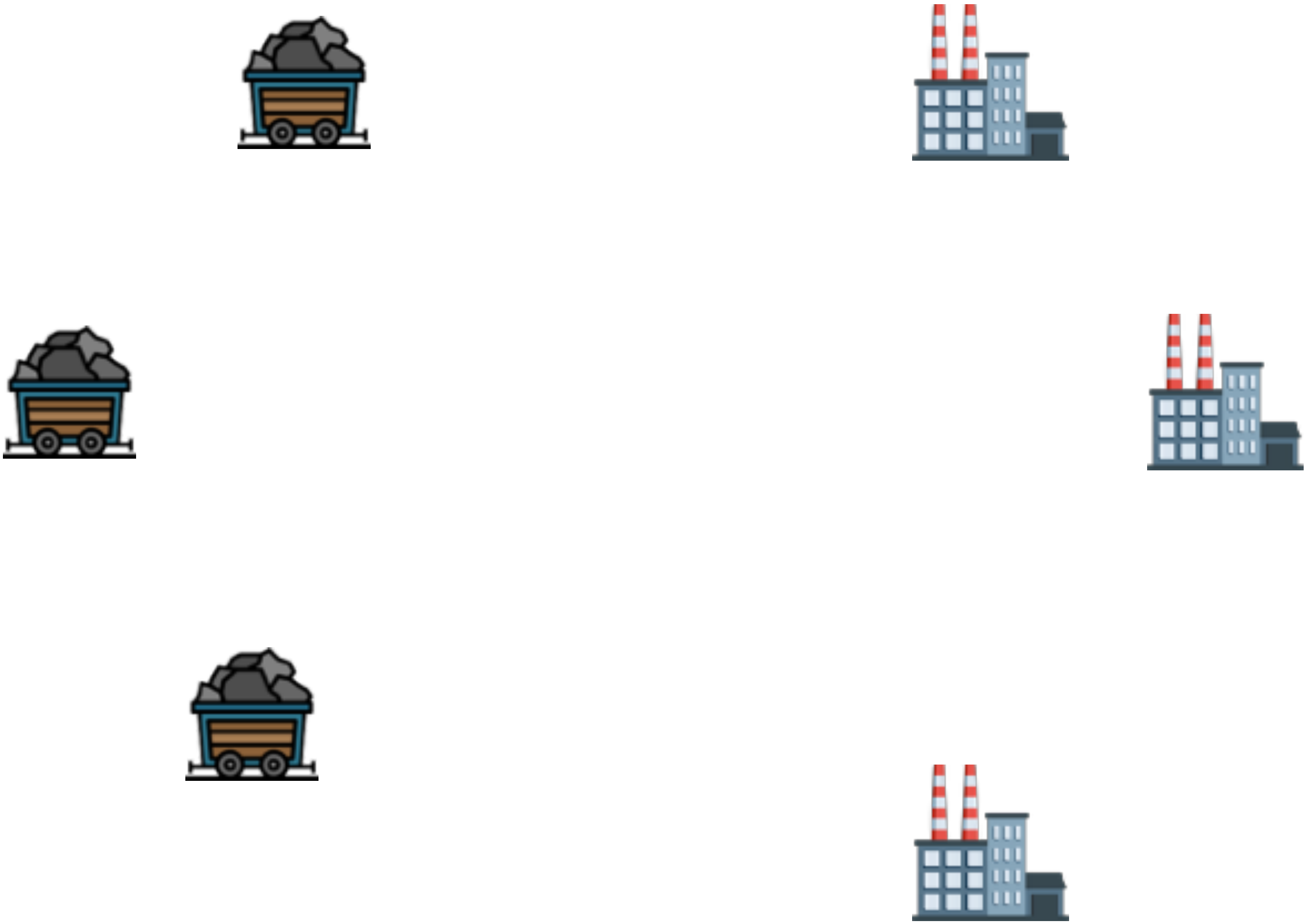
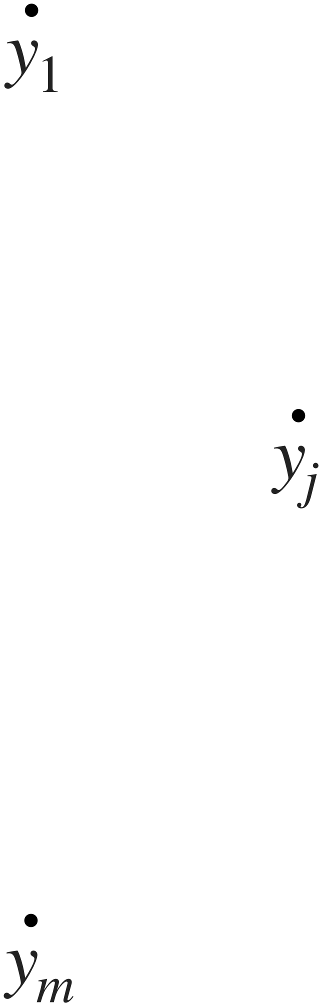


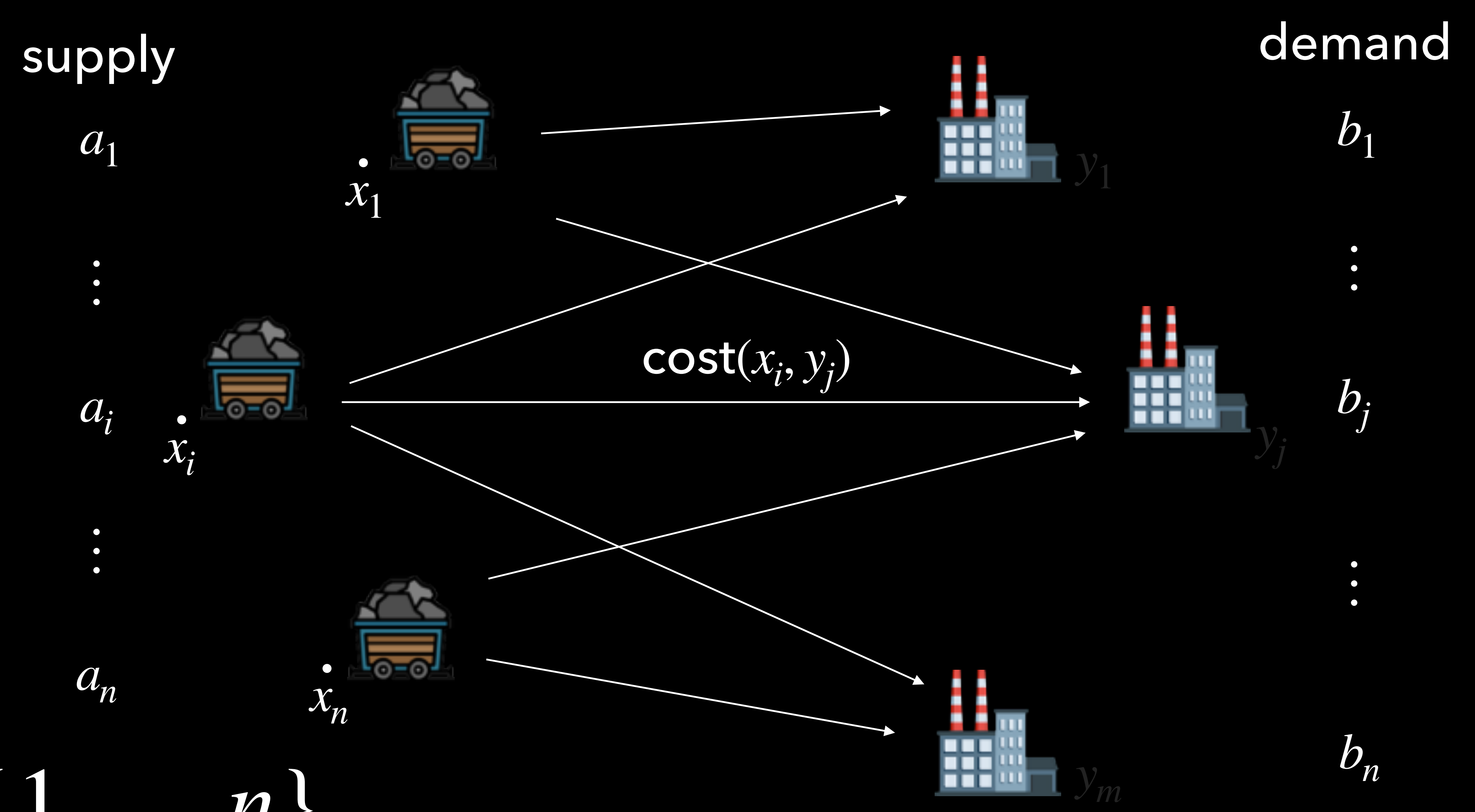

y_1
y_j
y_m

Monge's Problem:

(Matching/permutation version)

$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^{n} \mathbf{C}_{i,\sigma(i)}$$

Monge Formulation



Data indexed by
$i \in \{1,\ldots,n\}, j \in \{1,\ldots,n\}$

Cost matrix: $(C_{i,j})_{i,j=1}^{n} \in \mathbb{R}^{n\times n}$

Search over permutations in
$\Sigma_n = \{\sigma : \{1,\ldots,n\} \to \{1,\ldots,n\}\}$

We want the min-cost one

Monge's Problem:

(Matching/permutation version)

$$\min_{\sigma\in\Sigma_n} \frac{1}{n} \sum_{i=1}^{n} \mathrm{C}_{i,\sigma(i)}$$

Solving Monge's Problem

- We already mentioned how to solve the discrete version in general: Hungarian algorithm (sadly, $O(n^3)$)
- But there are settings where it can be solved faster!